***Conditional Entropy***

Let $x$ and $y$ be discrete random variables. Then the conditional entropy, $H(x \mid y)$, is the weighted average of the entropies of $x$ <u>given a possible value of</u> $y$, over all the different possible values of $y$. The weights of each entropy of $x$ are with respect to the probability of the possible value of $y$, $P(y)$.

If $x \in X$ and $y \in Y$ then $H(x \mid y)$ may be computed directly using:

$$H(x \mid y) = \sum_{y \in Y} P(y) \cdot \left[ -\sum_{x \in X} P(x \mid y) \cdot \log_2 \left[ P(x \mid y) \right] \right]$$

**Example1:**

Let $x \in \{000, 010, 100, 110, 001, 011, 101, 111\}$ represent the result of flipping a coin three times.

Let $y \in \{0,1\}$ represent the result of the <u>last</u> flip.

Find $H(x \mid y)$, the conditional entropy of $x$ given that the result of the last flip is known.

There are two possible values of $y$ to consider: $y = 0$ and $y = 1$.

When $y = 0$, then $x$ can only be $\{000, 010, 100, 110\}$ with $P(x) = \dfrac{1}{4}$ for each of these. Then:

$$\begin{aligned} H(x \mid y=0) &= -\frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] \\ &= -\frac{1}{4}(-2) - \frac{1}{4}(-2) - \frac{1}{4}(-2) - \frac{1}{4}(-2) \\ &= 2 \end{aligned}$$

When $y = 1$, then $x$ can only be $\{001, 011, 101, 111\}$ with $P(x) = \dfrac{1}{4}$ for each of these. Then:

$$\begin{aligned} H(x \mid y=1) &= -\frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] - \frac{1}{4}\log_2\left[\frac{1}{4}\right] \\ &= -\frac{1}{4}(-2) - \frac{1}{4}(-2) - \frac{1}{4}(-2) - \frac{1}{4}(-2) \\ &= 2 \end{aligned}$$

The conditional entropy of $x$ given that the result of the last flip is known is:

$$
\begin{aligned}
H(x \mid y) &= \sum_{y \in Y} P(y) \cdot \left[ -\sum_{x \in X} P(x \mid y) \cdot \log_2 \left[ P(x \mid y) \right] \right] \\
&= P(y=0) \cdot \left[ -\sum_{x \in X} P(x \mid y=0) \cdot \log_2 \left[ P(x \mid y=0) \right] \right] + P(y=1) \cdot \left[ -\sum_{x \in X} P(x \mid y=1) \cdot \log_2 \left[ P(x \mid y=1) \right] \right] \\
&= P(y=0) \cdot \left[ H(x \mid y=0) \right] + P(y=1) \cdot \left[ H(x \mid y=1) \right] \\
&= \frac{1}{2} \cdot \left[ H(x \mid y=0) \right] + \frac{1}{2} \cdot \left[ H(x \mid y=1) \right] \\
&= \frac{1}{2} \cdot \left[ 2 \right] + \frac{1}{2} \cdot \left[ 2 \right] \\
&= 2
\end{aligned}
$$

***Key Equivocation***

Consider a cryptosystem with plaintexts $x \in M$, ciphertexts $y \in C$, and keys $k \in K$.

The amount of remaining uncertainty in the key given that the ciphertext is known, $H(k \mid y)$, is called the key equivocation. It may be computed directly using:

$$H(k \mid y) = \sum_{y \in C} P(y) \cdot \left[ -\sum_{k \in K} P(k \mid y) \cdot \log_2 \left[ P(k \mid y) \right] \right]$$

The key equivocation may also be computed indirectly by the following theorem:

$$H(k \mid y) = H(k) + H(x) - H(y)$$

**Example2:**

A cryptosystem with plaintexts $x \in M = \{a, b\}$, ciphertexts $y \in C = \{1,2,3,4\}$, and keys $k \in K = \{k_1, k_2, k_3\}$ has encryption matrix:

| | *a* | *b* |
|---|---|---|
| $k_1$ | 1 | 2 |
| $k_2$ | 2 | 3 |
| $k_3$ | 3 | 4 |

And probability distributions: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

If the ciphertext is $y = 2$ then:

$$P(k = k_1 \mid y = 2) = \frac{P(k = k_1) \cdot P(y = 2 \mid k = k_1)}{P(y = 2)} = \frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{7}{16}} = \frac{6}{7}$$

$$P(k = k_2 \mid y = 2) = \frac{P(k = k_2) \cdot P(y = 2 \mid k = k_2)}{P(y = 2)} = \frac{\frac{1}{4} \cdot \frac{3}{4}}{\frac{7}{16}} = \frac{1}{7}$$

$$P(k = k_3 \mid y = 2) = \frac{P(k = k_3) \cdot P(y = 2 \mid k = k_3)}{P(y = 2)} = \frac{\frac{1}{4} \cdot 0}{\frac{7}{16}} = 0$$

Then the amount of remaining uncertainty in the key given that the ciphertext is $y=2$ is:

$$\begin{aligned}
-\sum_{k\in K} P(k\mid y=2)\cdot\log_2\left[P(k\mid y=2)\right] &= -\frac{6}{7}\log_2\left[\frac{6}{7}\right]-\frac{1}{7}\log_2\left[\frac{1}{7}\right]-0\\
&= -\frac{6}{7}(-0.22239...)-\frac{1}{7}(-2.80735...)-0\\
&\cong 0.59167...
\end{aligned}$$

To obtain the <u>key equivocation</u> we would compute the amount of remaining uncertainty in the key for other three cases where the ciphertext is $y=1$ or $y=3$ or $y=4$, and then sum all the cases according their probabilities:

$$\begin{aligned}
H(k\mid y) &= \sum_{y\in C} P(y)\cdot\left[-\sum_{k\in K} P(k\mid y)\cdot\log_2\left[P(k\mid y)\right]\right]\\
&= P(y=1)\times H(k\mid y=1)+P(y=2)\times H(k\mid y=2)+P(y=3)\times H(k\mid y=3)+P(y=4)\times H(k\mid y=4)\\
&= P(y=1)\times H(k\mid y=1)+\frac{7}{16}\times(0.59167...)+P(y=3)\times H(k\mid y=3)+P(y=4)\times H(k\mid y=4)
\end{aligned}$$

***Perfect Secrecy and Conditional Entropy***

Recall that a cryptosystem has <u>perfect secrecy</u> if $P(x\mid y)=P(x)$ for all $x\in M$ and $y\in C$.

Another way to determine perfect secrecy is to measure the uncertainty in the plaintext, $H(x)$, and compare this to the remaining uncertainty in the plaintext <u>given that</u> the ciphertext is known, $H(x\mid y)$. If they are equal, then knowing the ciphertext does not change the amount of information that we know about the plaintext.

Therefore, a cryptosystem has <u>perfect secrecy</u> if $H(x\mid y)=H(x)$.

$H(x\mid y)$ may be computed directly using the general formula for conditional entropy with $x\in M$ and $y\in C$:

$$H(x\mid y)=\sum_{y\in C} P(y)\cdot\left[-\sum_{x\in M} P(x\mid y)\cdot\log_2\left[P(x\mid y)\right]\right]$$

**Exercises:**

**1.** Let $x \in \{1,2,3,4,5,6\}$ represent the result of rolling a 6-sided dice. Let $y \in \{0,1\}$ represent the if the roll is odd ( $y=1$ if the roll is odd and 0 otherwise). Determine $H(x \mid y)$.

**2.** A cryptosystem with plaintexts $x \in M = \{a,b\}$, ciphertexts $y \in C = \{1,2,3,4\}$, and keys $k \in K = \{k_1,k_2,k_3\}$ has encryption matrix:

| | *a* | *b* |
|---|---|---|
| $k_1$ | 1 | 2 |
| $k_2$ | 2 | 3 |
| $k_3$ | 3 | 4 |

And probability distributions: $P(a)=\frac{1}{4}, P(b)=\frac{3}{4}$ and $P(k_1)=\frac{1}{2}, P(k_2)=\frac{1}{4}, P(k_3)=\frac{1}{4}$

a) Determine $H(k \mid y=1)$.
b) Determine $H(k \mid y=3)$.
c) Determine $H(k \mid y=4)$.
d) Determine the key equivocation directly.
e) Determine the key equivocation indirectly (using $H(k \mid y) = H(k) + H(x) - H(y)$ ).

**3.** A cryptosystem with plaintexts $x \in M = \{s,t\}$, ciphertexts $y \in C = \{0,1\}$, and keys $k \in K = \{k_1,k_2\}$ has encryption matrix:

| | *s* | *t* |
|---|---|---|
| $k_1$ | 1 | 0 |
| $k_2$ | 0 | 1 |

And probability distributions: $P(s)=0.3,\ P(t)=0.7$ and $P(k_1)=\frac{1}{2},\ P(k_2)=\frac{1}{2}$

a) Determine the key equivocation directly.
b) Determine the key equivocation indirectly.
c) Determine $H(x \mid y=0)$.
d) Determine $H(x \mid y=1)$.
e) Prove that this system has <u>perfect secrecy</u> by showing that $H(x \mid y) = H(x)$.

**Answers:**

**1.** $H(x \mid y) \cong 1.5850$

**2.**

a) $H(k \mid y=1) = 0$

b) $H(k \mid y=3) \cong 0.8113$

c) $H(k \mid y=4) = 0$

d) $H(k \mid y) = 0 + 0.25885... + 0.20281... + 0 \cong 0.4617$

e) $H(k \mid y) = 1.5 + 0.81127... - 1.84960... \cong 0.4617$

**3.**

a) $H(k \mid y) = 0.44064... + 0.44064... \cong 0.8813$

b) $H(k \mid y) = 1 + 0.88129... - 1... \cong 0.8813$

c) $H(x \mid y=0) \cong 0.8813$

d) $H(x \mid y=1) \cong 0.8813$

e) $H(x \mid y) = H(x) \cong 0.8813$